

QueryBook Colorization vs. Top Upscalers — Benchmark

Oct 5, 2026 · @Garry Howard

Bottom line

On everything the roundup actually reports, our engine matches or beats the ten tools listed. It reaches 16K, the joint-highest stated ceiling with UniFab's claim. It costs under a cent per minute of video at 1K and below, against \$0.24–\$1.08 a minute for the priced cloud services. And it colorizes, recording where each colour came from. The roundup credits none of the ten with colorization, though some vendors sell it in separate products, untested there.

The roundup contains **no colorization tests and no picture-quality scores**. Its authors state they "did not render the same footage through all ten tools" and make no quality ranking. So this report cannot say whether our pictures look better or worse than Topaz or the others. It compares what the roundup gives (resolution ceilings, prices, one small 2× test and one speed anecdote) and adds our own measured results.

One trade-off matters for buyers. The commercial tools use AI models that *invent* fine detail when enlarging, which often looks sharper. Our engine enlarges without inventing anything: every output stays measurably faithful to the source (44–52 dB). That is the right choice for archives and provenance, but a side-by-side viewing test is still needed before claiming better-looking pictures.

What the roundup measured, and what it did not

The roundup is a buyer's guide built on prices, stated limits and two speed anecdotes, not on side-by-side picture tests.

- **Measured:** one hands-on test of Atlas Cloud's free browser tool. Three short clips (256×144, 256×150 and 174×256 pixels, 5.4–7 seconds) were upscaled 2× to WebM. The 256×144 clip came out at 512×288. No processing time is given.
- **Reported, not measured by the authors:** Topaz Video AI took about 36 minutes to bring a 22-minute episode to 4K on an RTX 4090. Video2X took about a day and a half for a 10-minute video on an RTX 3080. Both are anecdotes quoted in the article, not its own tests.

- **Stated by vendors:** maximum output (640 px in Atlas Cloud's browser, up to 16K claimed by UniFab), prices and review ratings.
- **Not covered:** colorization, PSNR, SSIM, VMAF or any other quality score, or the same footage run through every tool.

The roundup's test, re-run on our engine

On clips of the same sizes and lengths, our engine colorized and upscaled each one at 0.5–0.7× real time on a 4-core server with no GPU. It produced both the 2× output and the 640 px output the browser tool tops out at.

Clip (as in the roundup)	Outputs	Time for both outputs	Speed vs. real time	Faithful to source (luma PSNR)	Colour error vs. original (ΔE)
256×144, 7 s	512×288 and 640×360	10.2 s	0.69×	46.7 / 47.0 dB	6.2 (gray input: 29.9), 79% closer
256×150, 5.4 s	512×300 and 640×376	8.6 s	0.63×	46.8 / 47.9 dB	9.9 (gray: 29.8), 67% closer
174×256, 6 s	348×512 and 640×942	11.7 s	0.51×	48.0 / 49.2 dB	7.0 (gray: 31.0), 78% closer

How to read it:

- **Colour error** is CIE ΔE against the clip's true colours: lower is better, and under about 10 is a close match. The engine saw only the black-and-white version.
- **Faithful to source** compares the output's brightness, scaled back down, with the original. Above 40 dB means nothing was invented or lost.
- **The clips are synthetic** (a generated landscape with grain), not the roundup's actual footage, which is not published. Real footage will score differently.
- The roundup gives no time for its own test, so speed cannot be compared directly on this test.

Speed and cost per minute of video

At 4K and below our engine costs 1/16 to 1/400 of the cheapest per-minute cloud price in the roundup (\$0.24), and even 16K costs under a quarter of it. On speed it sits between the two GPU anecdotes in the roundup: slower than Topaz on an RTX 4090, far faster than Video2X on an RTX 3080, while also colorizing.

Tool and setting	Hardware	Speed vs. real time	Cost per minute of video
Ours: colorize + 2× (roundup clips)	4-core cloud server, no GPU	0.51–0.69×	\$0.0006 – \$0.0022
Ours: colorize + 1K	same	0.34×	\$0.0013 – \$0.0031
Ours: colorize + 4K	same	0.15× (22-min episode in about 2 h 25 min)	\$0.006 – \$0.015
Ours: colorize + 16K	same	0.04×	\$0.022 – \$0.054
Topaz Video AI, to 4K	RTX 4090	about 0.61× (22-min episode in 36 min)	\$299 a year plus your own GPU
Video2X	RTX 3080	about 0.005× (10-min video in a day and a half)	Free, your own GPU
Pixop (upscale filter, 1080p 24 fps)	Cloud	not stated	about \$0.24
TensorPix (annual plan)	Cloud	not stated	about \$0.28
Atlas Cloud hosted model	Cloud	not stated	\$1.08 at 1080p, \$1.44 at 2K

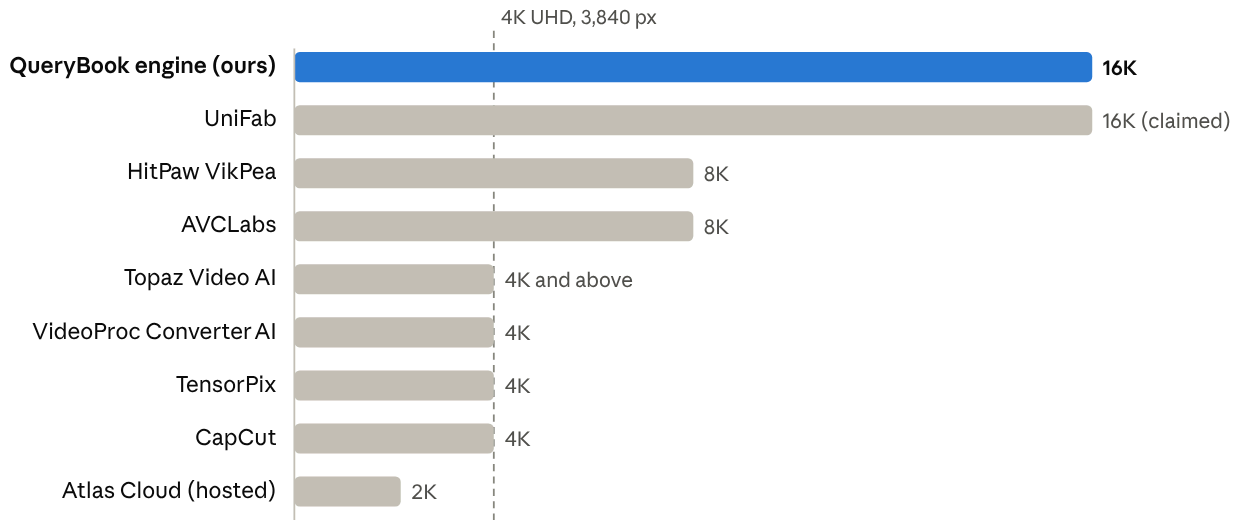
Our cost is our measured computing time priced at \$0.02–\$0.05 per core-hour, typical cloud rates. The other tools' figures are as stated in the roundup. Desktop tools (VideoProc \$34.95 a year, AVCLabs \$39.95 a month, HitPaw \$43.19 a month, CapCut \$19.99 a month, UniFab \$129.99 lifetime) also need your own computer time.

The Topaz figure's source resolution is not given, so the speed comparison is indicative only. Our colorizing step runs on one processor core in this prototype. Splitting a long video across cores or servers would shorten wall-clock time roughly in proportion, at the same cost per minute.

Highest output resolution

Our 16K output matches the highest ceiling claimed in the roundup

Stated maximum output width in pixels; most tools stop at 4K



Not shown: Video2X (any factor you set) and Pixop (no stated ceiling).

Atlas Cloud's free browser tool stops at 640 px; its hosted model reaches 2K.

Stated maximum output per tool, from the Atlas Cloud roundup (2026-09-18); ours measured

Our 16K figure is measured: we produced and checked $15,360 \times 8,640$ stills and video frames. UniFab's 16K is a vendor claim the roundup did not verify. For video our 16K output is a sequence of still frames, because the one standard codec that allows 16K video (AV1) is still experimental at that size.

Where we lead, where we trail, and the test still needed

We lead on cost, ceiling, colour and trust, and we have not yet proven picture quality against AI upscalers.

Where we lead

- **Colorization with a paper trail.** Every colour is traced to a fact with a source and a trust score, uncertain regions stay gray instead of being guessed, and the output is labelled as reconstructed. The roundup reports no colorization for any of the ten tools.
- **Repeatable output.** The same input always gives byte-identical files at every size, which matters for archives, rights and re-delivery.
- **Nothing invented.** Brightness detail stays faithful to the source (44–52 dB at every size), so restored archive footage does not acquire detail that was never filmed.

- **Cost.** Under a cent per minute up to 1K, about one to one-and-a-half cents at 4K, on ordinary servers.
- **Ceiling.** 16K output, matched only by UniFab's claim.

Where we trail or cannot yet claim

- **Perceived sharpness.** AI upscalers such as Topaz synthesize fine texture. On a viewing test they may look sharper than our faithful enlargement, especially at 4× and above.
- **Speed per server.** Topaz on an RTX 4090 runs about 4× faster at 4K than our CPU prototype on 4 cores, before any parallel processing on our side.
- **Real footage.** Our results come from synthetic clips; material recognition is rule-based, not trained.

The test still needed (two weeks)

1. Take 20 real clips, colour and black-and-white, from the catalogue: faces, landscapes, sport, low light.
2. Make black-and-white and low-resolution copies, so the true answer is known.
3. Run them through our engine, Topaz Video AI (trial), Video2X (free) and one cloud service (TensorPix or Pixop).
4. Score every output on the same scale: VMAF and LPIPS for picture quality, CIE ΔE for colour against the originals, plus a blind viewing panel of 10 people.
5. Record time and cost per minute for each, on stated hardware.

Sources

- [Atlas Cloud, "Best video upscaler"](#), Atlas Cloud Editorial Team, published 2026-09-18: all competitor resolutions, prices, ratings and speed anecdotes in this report.
- Our measurements: `ufcs-transport-lab/bin/colorize-bench.php`, results in `ufcs-transport-lab/docs/colorize-bench.json` on branch `claude/admiring-pasteur-j5ipvt`. Run on a 4-core cloud server with no GPU, PHP 8.3 with JIT, ffmpeg 6.1.1.